

ORC · LONG-RANGE CASIMIR SERIES · PAPER II

Gravitational Gradient Backgrounds in Long-Range Casimir Measurements

A Tidal-Tensor Noise Budget for the Sub-Millimeter Regime

Tidal force $\sim 10^{-13}$ N at unshielded 1 m · Safety perimeter scaling · Falsifiable brick-test with dummy-mass control

Identifies when nearby moving masses become a relevant systematic for $d > 100 \mu\text{m}$.

D. Kadunic · Omega Research Collective · May 2026

doi:10.5281/zenodo.1082226

Paper II of three. Patches (Paper I) treated; thermal-mechanical (Paper III) follows.

On the question: when the Casimir signal is 10^{-17} N, who is allowed to walk past the lab door?

ABSTRACT

We present a tidal-tensor noise budget for gravitational-gradient backgrounds in long-range Casimir force measurements. For a 1 cm^2 parallel-plate geometry at $d = 1 \text{ mm}$ separation with thermal Casimir signal $F_T \approx 4.0 \times 10^{-17} \text{ N}$, a 70 kg person at 1 m produces an upper-bound tidal force of $|\delta F|_{\text{max}} \approx 8.4 \times 10^{-14} \text{ N}$ — approximately 2,100 times the Casimir signal under worst-case unshielded coupling. The minimum exclusion distance for an unshielded apparatus to bring tidal noise below thermal Casimir is $R_{\text{min}} \approx 12.8 \text{ m}$ for a person and 6.7 m for a 10 kg laboratory object; for the zero-point Casimir contribution $F_0 \approx 1.3 \times 10^{-19} \text{ N}$, the perimeter extends to $\sim 86 \text{ m}$. **These are upper bounds for unshielded, perfectly-aligned, point-source coupling;** realistic shielded experiments achieve coupling reductions of 1–3 orders of magnitude through differential geometry, vibration isolation, and background subtraction. Past sub-micron experiments (Decca 2005/2007, Sushkov 2011) operated at separations where Casimir dominated gradient backgrounds by many orders of magnitude and are *not* threatened by this analysis. The relevance is to future long-range claims at $d > 100 \mu\text{m}$, where the Casimir signal falls into the same order of magnitude as unshielded gradient backgrounds and active control becomes essential. We propose a falsifiable brick-test protocol with required inert dummy-mass control: a 10 kg non-magnetic mass moved between $R = 0.5, 1.0, 2.0 \text{ m}$ should produce force changes of $9.6 \times 10^{-14}, 1.2 \times 10^{-14}, 1.5 \times 10^{-15} \text{ N}$ respectively, distinguishable above the thermal Casimir floor at the smaller distances. The required dummy-mass control eliminates vibration and electrostatic confounds that would otherwise mimic gravitational coupling.

1. Introduction and Scope

Paper I of this series treated electrostatic patch potentials, identifying the $d \sim 3\text{--}10 \mu\text{m}$ sphere-plane window as the danger zone for that systematic. This paper treats the next noise domain: gravitational-gradient backgrounds from nearby moving masses. The relevance scales differently from patches: gradient backgrounds are independent of d (they depend on the

source mass and distance, not on the plate separation), but the Casimir signal falls as d^{-4} (zero-point) or d^{-3} (thermal), so gradient-to-Casimir ratios become significant only at long separations.

Scope and limitations. This paper analyzes only gravitational gradients. We do not treat tilt coupling, seismic-mechanical coupling, or coupled multi-systematic identifiability — those depend on the specific torsion-balance or MEMS implementation and require apparatus-specific noise modeling. We confine ourselves to a single clean question: at what range and for what masses do gravitational gradients from external bodies become comparable to the thermal Casimir signal in the long-range regime? The answer scales the apparatus design perimeter for any future long-range Casimir claim.

Past experiments are not contested. Decca et al. (sub-micron, sphere-plane) and Sushkov et al. (cryogenic, large-radius) operated at separations where the Casimir signal was 10^{-10} to 10^{-12} N — three to six orders of magnitude above any plausible unshielded gradient background. The analysis presented here is relevant to *future* claims at $d > 100 \mu\text{m}$, where the signal floor approaches the gradient background under unshielded conditions.

2. The Tidal Tensor Formalism

The gravitational acceleration from a point mass M at position $\mathbf{R}$ is $\mathbf{a}(\mathbf{r}) = -GM(\mathbf{r}-\mathbf{R})/|\mathbf{r}-\mathbf{R}|^3$. The leading differential coupling between two test masses separated by displacement δ is given by the tidal tensor

$$T_{ij} = (GM/R^3)(3 n_i n_j - \delta_{ij}), \quad \mathbf{n} = \mathbf{R}/R.$$

The differential force on a test mass m separated by δ is $F_{\text{grad}} = m \delta^T \mathbf{T} \mathbf{e} + O(\delta^2/R^4)$, where $\mathbf{e}$ is the sensitive-axis direction. The tidal tensor's largest eigenvalue is $2GM/R^3$, giving a maximum coupling

$$|\delta F|_{\text{max}} = 2 m G M \delta / R^3$$

for ideal axial alignment. For a finite-size source, the point-mass approximation *overestimates* the gradient by a correction factor of $O(L^2/R^2)$ where L is the source's size; for a person at $R = 1$ m, the overestimate is approximately 10–20%. This conservative bias is appropriate for safety-perimeter mapping.

3. Numerical Estimates for a Reference Apparatus

We take a reference apparatus consisting of two square copper plates of area $A = 1 \text{ cm}^2$, thickness $t = 1 \text{ mm}$, plate mass $m = \rho_{\text{Cu}} \cdot A \cdot t = 8960 \times 10^{-4} \times 10^{-3} = 8.96 \times 10^{-4} \text{ kg} = 0.896 \text{ g}$. The plates are separated by $d = 1 \text{ mm}$, with characteristic differential displacement $\delta = 1 \text{ cm}$ (the plate dimension) for tidal coupling.

3.1 Casimir Reference Signal

From Paper I (verified), the Casimir contributions at this geometry and $d = 1 \text{ mm}$ are:

$$F_T = 3.96 \times 10^{-17} \text{ N (thermal, } T = 300 \text{ K)},$$

$$F_0 = 1.30 \times 10^{-19} \text{ N (zero-point)}.$$

3.2 Tidal Force from Common Sources

Applying $|\delta F|_{\text{max}} = 2 m G M \delta / R^3$ with the reference apparatus:

Source	M	R	$ \delta F _{\text{max}}$ (N)	Ratio to F_T
--------	---	---	-------------------------------	----------------

Adult experimenter	70 kg	1 m	8.4×10^{-14}	2,100
Adult experimenter	70 kg	5 m	6.7×10^{-16}	17
Adult experimenter	70 kg	12.8 m	4.0×10^{-17}	1.0
Lead brick (10 kg)	10 kg	1 m	1.2×10^{-14}	300
Lead brick (10 kg)	10 kg	0.5 m	9.6×10^{-14}	2,400
Lab equipment (1 kg)	1 kg	1 m	1.2×10^{-15}	30
Mobile phone (0.1 kg)	0.1 kg	1 m	1.2×10^{-16}	3

Headline finding: in the worst-case unshielded configuration, common laboratory objects produce gravitational backgrounds 10–10,000× larger than the thermal Casimir signal at $d = 1$ mm. This sets a strict requirement on either physical exclusion (Section 4), active shielding, or differential geometry that cancels common-mode gradient contributions.

4. Safety Perimeter Map

Inverting the tidal expression, the minimum exclusion distance for an unshielded configuration to bring $|\delta F|_{\max}$ below a target threshold F_{thr} is

$$R_{\min} = (2 m G M \delta / F_{\text{thr}})^{1/3}.$$

Table 2 reports $R_{\min}$ for the thermal Casimir threshold $F_T = 4.0 \times 10^{-17}$ N for a range of source masses.

Source mass	Example	$R_{\min}$ for F_T (m)	$R_{\min}$ for F_0 (m)
70 kg	Adult experimenter	12.8	86
10 kg	Lead brick / large equipment	6.7	45
1 kg	Small instrument	3.1	21
0.1 kg	Phone / hand tool	1.5	9.7

Important caveats. These distances are upper-bound exclusion radii under three pessimistic assumptions: (i) the source is treated as a point mass, ignoring extended-body averaging that reduces the gradient by 10–20% at the relevant geometries; (ii) ideal axial alignment of the tidal tensor's largest eigenvalue with the apparatus's sensitive axis, which is rarely realized in practice; (iii) no differential-mode cancellation by torsion-balance or sphere-plane geometry, which provides 1–3 orders of magnitude common-mode rejection in real experiments.

Realistic shielded experiments therefore tolerate sources at $R_{\min}/3$ to $R_{\min}/10$. The values in Table 2 are appropriate as *conservative design perimeters* for facility planning (where to allow personnel access during a measurement run, what masses to permit in adjacent rooms), not as fundamental impossibility bounds.

5. Why Past Sub-Micron Experiments Are Unaffected

The gradient-to-Casimir ratio scales as $F_{\text{grad}}/F_C \propto d^3$ (for thermal Casimir, since $F_C \propto 1/d^3$) or d^4 (for zero-point). At sub-micron separations, the ratio is much smaller than at millimeter separations:

Experiment	d	F_C (rough)	F_{grad} (person, 1m)	Ratio
Decca 2005/2007	750 nm	$\sim 2 \times 10^{-10}$ N	8×10^{-14} N	4×10^{-4}
Sushkov 2011	2 μ m	$\sim 5 \times 10^{-12}$ N	8×10^{-14} N	1.6×10^{-2}

Hypothetical (this paper)	1 mm	4×10^{-17} N	8×10^{-14} N	2,100
---------------------------	------	-----------------------	-----------------------	-------

Decca and Sushkov's measurements were not gradient-limited. The Casimir signal was 10^3 – 10^4 times larger than any plausible unshielded gradient background at their separations. This analysis does not contest those results. It identifies the parameter regime where future experiments need to take on additional shielding burdens.

6. A Falsifiable Brick Test

The tidal scaling $1/R^3$ is a strong, characteristic signature: a factor-of-2 change in distance produces a factor-of-8 change in force. Vibrational, electrostatic, and thermal-drift confounds do not generically follow this scaling. A measurement of force change versus the position of a known external mass therefore directly tests the gradient-coupling model.

6.1 Protocol

Step	Action
1	Establish baseline force at $d = 1$ mm for 10 minutes (apparatus at rest)
2	Place a 10 kg non-magnetic mass (lead, brass, copper) at $R = 2.00$ m; measure for 5 min
3	Move mass to $R = 1.00$ m; measure for 5 min
4	Move mass to $R = 0.50$ m; measure for 5 min
5	Repeat steps 2–4 with electrostatically/dimensionally identical inert dummy mass
6	Differential subtract: $\Delta F_{\text{grav}} = \Delta F(\text{real mass}) - \Delta F(\text{dummy mass})$

The dummy-mass control is essential. Moving any object near the apparatus introduces vibrations, displaces air, and may carry adventitious surface charges. A mass that has the same dimensions, surface coating, and motion path as the real mass — but with negligible inertial mass (e.g., hollow with the same outer dimensions, or made of foam metallized to match the surface) — controls all of these confounds. The differential subtraction isolates the gravitational signal.

6.2 Predictions and Decision Criteria

R	Predicted ΔF_{grav}	Ratio to F_T	Decision
0.5 m	9.6×10^{-14} N	2,400	Strong signal expected
1.0 m	1.2×10^{-14} N	300	Robust signal expected
2.0 m	1.5×10^{-15} N	38	Marginal but detectable
Scaling test	$1/R^3$ ratio = 8:1	—	0.5 m / 1.0 m ratio

Decision criterion. The ratio of measured ΔF at $R = 0.5$ m to ΔF at $R = 1.0$ m should equal $8 \pm$ measurement uncertainty under pure tidal coupling. Deviations from $1/R^3$ scaling indicate either residual systematics or a non-gravitational coupling mechanism. The test is independent of the absolute force calibration: it relies only on the geometric ratio.

Magnetic-safety clause. The mass must be non-magnetic to avoid $1/R^3$ magnetic dipole interactions which would mimic gradients. Lead, brass, copper, or non-ferromagnetic alloys are appropriate. The dummy mass should match the magnetic susceptibility of the real mass to within standard tolerances.

Air-mass correction. A 10 kg mass displaces ~ 1 liter of air (density ~ 1.2 g/m³, so $\sim 10^{-3}$ kg displaced). The corresponding gradient correction is $< 10^{-17}$ N, comparable to F_T but well below the predicted tidal signals at $R \leq 1$ m. Negligible.

7. Brick-Test Error Budget

Parameter	Value	Uncertainty	Effect on ΔF
Source mass M	10.00 kg	± 0.10 kg	$\pm 1\%$
Distance R	0.500–2.000 m	± 0.005 m	$\pm 3\%$
Plate mass m	0.896 g	± 0.005 g	$\pm 0.5\%$
Differential δ	1.0 cm	± 0.05 cm	$\pm 5\%$
Newtonian model	—	$< 1\%$	Negligible
Total systematic	—	—	$\sim 6\%$

Statistical noise from a 5-minute integration at typical force-sensor resolution (10^{-15} N/ $\sqrt{\text{Hz}}$) is approximately 6×10^{-17} N — comparable to F_T , well below the predicted tidal signals at $R \leq 1$ m. The brick test is therefore robust at the larger predicted signal levels and provides a clean discrimination of the gradient hypothesis.

8. Limitations and Scope

(i) This paper analyzes only gravitational-gradient backgrounds from external sources. Tilt coupling, seismic-mechanical coupling, and thermal-drift contributions are deferred to Paper III, where they are treated with apparatus-specific transfer-function modeling. **(ii)** Point-mass and full-eigenvalue-alignment assumptions overestimate gradient coupling by factors of 1.2–10 in realistic configurations; the safety-perimeter values should be treated as conservative design bounds, not impossibility limits. **(iii)** Differential and common-mode rejection in sphere-plane and torsion-balance geometries provides 1–3 orders of magnitude additional suppression beyond the point-source bound, depending on apparatus symmetry. **(iv)** We treat tidal-tensor coupling at leading order in δ/R ; second-order corrections become relevant for source-apparatus distances comparable to apparatus dimensions.

9. Conclusions

Gravitational-gradient backgrounds from nearby moving masses are a relevant noise budget term for long-range Casimir measurements at $d > 100$ μm . For a reference 1 cm^2 apparatus at $d = 1$ mm, an unshielded experimenter at 1 m distance produces tidal forces of order 10^{-13} N — approximately 2,100 times the thermal Casimir signal. The conservative safety perimeter for unshielded operation extends to ~ 13 m for personnel and ~ 7 m for 10 kg laboratory objects. Realistic shielded configurations achieve coupling reductions of 1–3 orders of magnitude, but explicit gradient-control measures become a necessary part of the experimental protocol at long range.

Past sub-micron Casimir experiments are not threatened by this analysis. Decca (2005/2007) and Sushkov (2011) operated at separations where the Casimir signal exceeded any plausible unshielded gradient background by 3–4 orders of magnitude. The relevance is to *future* experiments venturing into the $d > 100$ μm regime, where such shielding becomes essential.

The brick test with required dummy-mass control provides a clean, falsifiable discrimination of gravitational coupling from electrostatic, vibrational, and thermal confounds. The $1/R^3$ scaling between $R = 0.5$ m and $R = 1.0$ m gives an 8:1 force ratio independent of absolute calibration. We recommend it as a standard verification step for any future long-range Casimir claim.

Paper III of the Long-Range Casimir Series treats thermal-mechanical mode coupling and tilt-induced parallelism systematics, completing the noise-budget program for the regime.

Acknowledgments

This paper underwent multiple rounds of red-team review across the Omega Research Collective. Earlier drafts contained framework-level claims (Fisher Information Collapse, identifiability theorems) that did not survive independent audit, along with a labeled-but-incorrect self-gravity calculation and 20% prefactor errors in the brick-test predictions. The final audit pass scoped the paper to its defensible content: a tidal-tensor noise budget for gravitational-gradient backgrounds and a falsifiable brick test with dummy-mass control. We thank the collective for their iterative work; the discipline that produced four rounds of corrections is the same discipline that produces the central scientific claim. Errors caught are physics learned, and a shorter honest paper is worth more than a long elaborate one that does not survive scrutiny.

References

- [1] Casimir H.B.G., 1948, *Proc. K. Ned. Akad. Wet.* **51**, 793.
- [2] Decca R.S. et al., 2005, *Phys. Rev. D* **71**, 083506; 2007, *Phys. Rev. D* **75**, 077101.
- [3] Sushkov A.O., Kim W.J., Dalvit D.A.R., Lamoreaux S.K., 2011, *Phys. Rev. Lett.* **107**, 171101.
- [4] Klimchitskaya G.L., Mohideen U., Mostepanenko V.M., 2009, *Rev. Mod. Phys.* **81**, 1827.
- [5] Misner C.W., Thorne K.S., Wheeler J.A., 1973, *Gravitation*, Freeman. (Tidal-tensor formalism)
- [6] Westphal T., Hepach H., Pfaff J., Aspelmeyer M., 2021, *Nature* **591**, 225. Reference for sub-mg-scale gravitational coupling measurement and contemporary tidal-noise control techniques.
- [7] Adelberger E.G., Heckel B.R., Nelson A.E., 2003, *Annu. Rev. Nucl. Part. Sci.* **53**, 77. Tests of the gravitational inverse-square law (gradient-control techniques).
- [8] Kadunic D., 2026, ORC Long-Range Casimir Series Paper I (companion paper, electrostatic patches).
- [9] Kadunic D., 2026, ORC Long-Range Casimir Series Paper III (forthcoming, thermal-mechanical mode coupling).